

CSC5041 Introduction to Databases

Teaching Plan — AY2026–27, Term 1

Lecturer: Prof. Lin Xuemin

Tutors: Zeng Ziyi (Tutorials 1, 3, 5) • Chen Junan (Tutorials 2, 4, 6)

Weekly schedule

Week	Date	Lecture topic	Tutorial	Tutor
1	9 Sep	Introduction to Databases: database systems, DBMS architecture, three-schema architecture, data independence, course overview	—	—
2	16 Sep	Conceptual Database Design: the Entity-Relationship model, entities, attributes, relationships, cardinality and participation constraints	—	—
3	23 Sep	The Relational Data Model and ER-to-relational mapping, integrity constraints	Tutorial 1	Zeng Ziyi
4	30 Sep	Relational Algebra: selection, projection, set operators, joins, division	—	—
5	7 Oct [†]	SQL, Part I: SELECT-FROM-WHERE, sorting, aggregation, grouping, nested queries	Tutorial 2	Chen Junan
6	14 Oct	SQL, Part II: advanced and correlated subqueries, set comparison; introduction to PL/pgSQL In-class Test 1 (covers Weeks 1–5)	—	—
7	21 Oct	Functional Dependencies: inference rules, attribute closure, candidate keys, minimal cover; Normal Forms 1NF–BCNF	Tutorial 3	Zeng Ziyi
8	28 Oct	Relational Database Design I: decompositions, lossless-join and dependency preservation	—	—
A1	TBA	Assignment 1 due	—	—
9	4 Nov	Relational Database Design II: 3NF and BCNF design algorithms	Tutorial 4	Chen Junan
10	11 Nov	Disk, Files and Buffer Management: storage hierarchy, file organisations, buffer replacement policies	—	—
11	18 Nov	Index Structures: ISAM, B ⁺ -trees, extendible hashing	Tutorial 5	Zeng Ziyi
12	25 Nov	Transaction Management I: ACID properties, schedules, conflict serialisability, precedence graphs In-class Test 2 (covers Weeks 6–11)	—	—
13	2 Dec	Transaction Management II: two-phase locking, timestamp ordering, deadlock handling, crash recovery	Tutorial 6	Chen Junan
14	9 Dec	Graph Databases and Big Graph Processing (overview only); course review	—	—
A2	TBA	Assignment 2 due	—	—
Exam	TBA	Final Examination	—	—

[†] 7 October falls within the National Day holiday (1–7 October 2026). The date shown is the scheduled Wednesday; this lecture and Tutorial 2 may be moved. A revised date will be announced if so.

In-class tests are held during the lecture slot of the week shown; assignments are submitted online by the due date shown.

Tutorial overview

Tutorial	Week	Date	Tutor	Theme
1	3	23 Sep	Zeng Ziyi	Conceptual and logical database design: ER modelling and mapping to relations
2	5	7 Oct [†]	Chen Junan	Relational algebra and basic SQL
3	7	21 Oct	Zeng Ziyi	Advanced SQL and functional dependencies
4	9	4 Nov	Chen Junan	Normalisation and relational database design
5	11	18 Nov	Zeng Ziyi	Storage, buffer management and index structures
6	13	2 Dec	Chen Junan	Transaction management

[†] 7 October falls within the National Day holiday; this tutorial may be moved.

Tutorial 1

Week 3

23 Sep

Conceptual and logical database design

Tutor: Zeng Ziyi

Lectures covered: Weeks 1–3

Topics

- Entities, attributes and keys; weak entities.
- Cardinality and participation constraints.
- ER-to-relational mapping: primary keys, foreign keys, and when a relationship needs its own table.

Tutorial 2

Week 5

7 Oct[†]

Relational algebra and basic SQL

Tutor: Chen Junan

Lectures covered: Weeks 4–5

Topics

- Relational algebra: select, project, join, set difference, division.
- SQL: GROUP BY, HAVING, NOT EXISTS, and NOT IN with NULLs.

Tutorial 3

Week 7

21 Oct

Advanced SQL and functional dependencies

Tutor: Zeng Ziyi

Lectures covered: Weeks 6–7

Topics

- Outer joins and COUNT(DISTINCT ...).
- Attribute closure and candidate keys.
- Minimal cover.

Tutorial 4	Week 9	4 Nov
Normalisation and relational database design		
Tutor: Chen Junan	Lectures covered:	Weeks 8–9

Topics

- Candidate keys and the highest normal form.
- Minimal cover; lossless-join and dependency-preserving 3NF decomposition.
- BCNF decomposition and dependency preservation.

Tutorial 5	Week 11	18 Nov
Storage, buffer management and index structures		
Tutor: Zeng Ziyi	Lectures covered:	Weeks 10–11

Topics

- Buffer replacement: LRU, MRU and FIFO compared by hit rate.
- B⁺-tree insertion and node splitting.

Tutorial 6	Week 13	2 Dec
Transaction management		
Tutor: Chen Junan	Lectures covered:	Weeks 12–13

Topics

- Conflicts and the precedence graph.
- Two-phase locking and strict 2PL.
- Recoverable, cascadeless and strict schedules.
- Recovery with write-ahead logging and checkpoints: redo and undo.
- Deadlock: detection and prevention.